Optimizing Chain-of-Thought Length in Small Language Models via Reinforcement Learning

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Overview

This project investigates how the length of chain-of-thought reasoning affects task performance in small language models, particulary Qwen2-0.5B-Instruct and Qwen2-1.5B-Instruct. It is composed of two major parts. Initially, an attempt is made to identify the empirically optimal reasoning length through controlled experiments. The experiments tested two differnt approaches of determining the optimal reasoning length, which were then compared. First one utilized beam search with differing length penalties in aim to observe a convergence to an optimal CoT length. The second drew multiple samples per problem under temperature sampling, which naturally varied the CoT length, and tested whether accuracy peaks at an intermediate reasoning length that — as suggested in *When More is Less: Understanding Chain-of-Thought Length in LLMs* (Wu et al., 2025) — balances performance against resource usage.

Then, reinforncement learning method - GRPO Shao et al. (2024) with explicit length regularization based on CoT length - was applied to shape model behaviour. We evaluated accuracy as well as the resulting CoT length across MATH Hendrycks et al. (2021) difficulty levels, and found that length-regularized GRPO improves accuracy while shortening the reasoning chains, most visibly on the harder problems.

1 Optimal CoT length. Empirical analysis

1.1 Approach 1: Beam search with length penalty

Our first approach manipulates CoT length at decoding time, leaving the model and the prompt fixed. We use beam search (k=4) and vary its length-normalization parameter α, which re-ranks completed beams by their log-probability divided by (length)$^{\alpha}$: negative α biases the search toward shorter chains, positive α toward more verbose ones. This gives a controlled way to vary CoT length without the two problems of prompt-based steering: the model following length instructions only imperfectly, and brevity instructions also changing the reasoning strategy. We sweep $\alpha \in \{-2, -1, 0, 0.5, 1, 2, 4\}$ on `Qwen2.5-0.5B-Instruct` over 20 MATH problems per difficulty level (levels 3–5), recording accuracy and output length to trace how the two co-vary as decoding is pushed shorter or longer.

Results Figure 1 shows that the length penalty had only a weak effect on the realized CoT length: the mean output grew slowly and monotonically from about 445 tokens at $\alpha = -2$ to about 470 at α=4. Accuracy stayed flat ($\approx 18.3\%$) for $\alpha \leq 0$, peaked at 20% for moderate positive penalties ($\alpha \in \{0.5, 1.0\}$), and then fell off for the most verbose setting, down to $\approx 16.7\%$ at α=4. This is consistent with the inverted-U hypothesis. Both very short and the longest chains underperform an intermediate length but the effect is small and the confidence bands are wide. This limited control motivated the within-problem sampling of Approach 2.

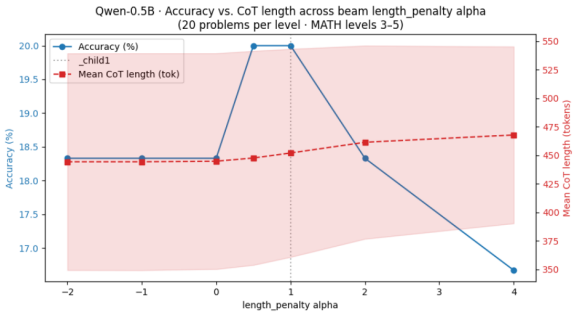
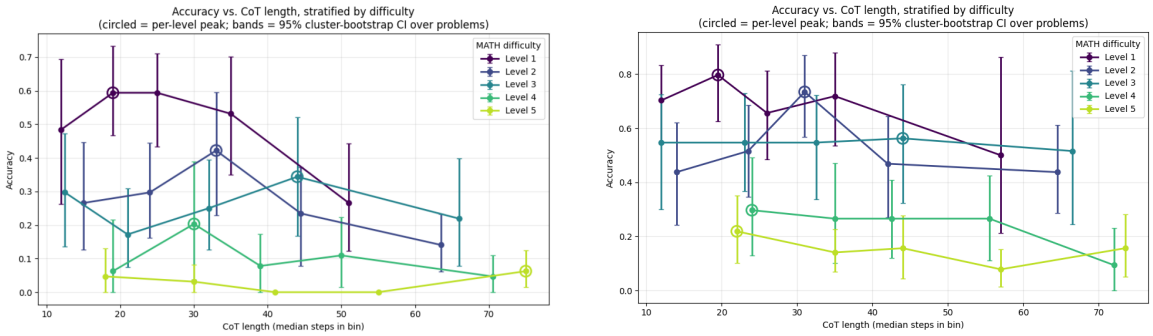


Figure 1: Accuracy vs. Chain of Thought length in beam search analysis

1.2 Approach 2: Within problem sampling



(a) Qwen 0.5B

(b) Qwen 1.5B

Figure 2: Accuracy vs. Chain of Thought length stratified by difficulty.

Wu et al. claimed that Chain of Thought reasoning exhibits a U-inverted relationship between length and accuracy. They did not focus on token cost, but defined CoT length by the muber of steps the model took to solve the problem. Too few reasoning steps under-determine the answer, while excessively long chains accumulate error, so accuracy peaks at some intermediate length and the optimum grows with problem difficulty and decreases with model length.

To test this hypothesis and determine optimal CoT length, simply grouping the solutions by length and comparing accuracy across different problems conflated two aspects:

1. Models emit longer chains on harder problems, which are also the problems they are more likely to get wrong. A negative length–accuracy correlation can therefore happen purely because of the reverse causation
2. Manipulating length by steering the prompt ("use exactly N steps") may fail since models comply imperfectly to the requirements. Moreover, instructed brevity may alter not only the length, but also the reasoning strategy.

Approach Within-problem sampling removes the difficulty confound by conditioning on the problem: for each problem (stratified across MATH levels 1–5), we drew completions under temperature sampling. This sampling alone allowed for varied CoT length across the draws, with no length instruction and no need for truncating the answers. We ranked each problem's draws (relative to its own draws) and pooled accuracy by rank (to compare lengths with difficulty fixed) and then binned samples by absolute length per level to test whether the accuracy peak shifts rightward as difficulty rises.

Results As it can be observed in Figure 2, for both model sizes (0.5B vs. 1.5B) and levels 1-4 of MATH dataset problems, there is a clear trend of accuracy peaking at an optimal number of steps and decreasing for longer reasonings. The easier the problems, the smaller the optimal CoT length is. Since Qwen 0.5B is a relatively small model, it may be that for very hard tasks (level 5) the optimal CoT would lie further to the right, although for Qwen-1.5B a plateou with no clear peak accuracies can also be observed. This still complies with the hypothesis that extended CoT length does not guarantee better model performance.

2 GRPO with length regularization

Recent research(Wu et al., 2025) has shown, that GRPO with binary correctness reward on math reasoning problems, exhibits simplicity bias, compressing the reasoning rollouts to optimal COT length. Motivated by this findings, we have decided to explore how a reward with implicit length regularization influences a behaviour of a small reasoning model.

2.1 Suggested reward formulation

Initially our reward was defined as follows:

$$R = R_{\text{correctness}} - \lambda \cdot \text{COTlength}$$

Where the COT length was measured in tokens (normalized).
Despite achieving measurable decrease in reasoning length on GSM8K Cobbe et al. (2021), we found that this reward was extremely sensitive to the choice of combination of length regularization strength λ and learning rate. For several runs we've observed complete collapse of the model, such that it was producing empty responses. We also couldn't scale this method to dataset of more challenging problems, namely MATH.

2.2 Proposed reward

We propose slightly modified definition of the reward function:

$$R = R_{correctness}(1 - \lambda \cdot \text{COTlength})$$

This approach essentially disables length regularization for incorrect responses. This obviously decreases model tendencies to collapse, hence makes the framework less sensitive to hyperparameter selection.
It also allowed us to apply this framework on more challenging dataset. Since for tough problems a small model is more frequently incorrect than correct, for the initial formulation the reward is very noisy.

2.3 Experimental Setup

Model. We use `Qwen2.5-0.5B-Instruct` as the base policy, a 0.5B-parameter instruction-tuned model. We fine-tune it with parameter-efficient LoRA adapters (rank r=16, α=32, dropout 0.05) attached to the attention and MLP projections (`q,k,v,o_proj` and `gate,up,down_proj`); the base weights remain frozen.

Data. Training uses the MATH dataset (`EleutherAI/hendrycks_math`), pooling *all seven subjects* (algebra, counting & probability, geometry, intermediate algebra, number theory, prealgebra, precalculus) across all five difficulty levels, for a total of 7,498 training problems.

Reinforcement learning. We optimize the policy with Group Relative Policy Optimization (GRPO; TRL 1.5.1), sampling G=8 completions per prompt. Since we wanted to benchamrk against

(Wu et al., 2025), we've decided to follow their proposed definition of COT length, measured by number of non-empty newline-separated lines. Additionaly, we divided the actuall step count by 32, to get roughly normalized length.

Training hyperparameters. We set $\lambda = 0.5$. Per-device batch size 2 with gradient accumulation 16 gives an effective batch of 32. We train for 300 steps with learning rate 1e−5, sampling temperature 1.0 and maximum completion length 768 tokens, seed 42. Training runs on a single GPU; as the available cards lack `bf16` support we use `fp16`.

Evaluation We define **optimal CoT length of a single problem** as an average CoT length of correct responses. We also introduce model specific **task difficulty** as a fraction of incorrect responses.
To compute both metrics we prompt the model to give n responses to each problem from evaluation set. We study how the relation between those two metrics changes under length regularized GRPO. We also study accuracy of model per each difficulty level of MATH dataset. We compare the tuned model against baseline and unregularized GRPO

We evaluate on MATH (algebra subject, test split, per difficulty level L1–L5), disjoint from the GRPO training data. Per level we draw a fixed seeded subset of 20 problems and, following the complexity-based prompting protocol of Wu et al. (2025), resample each problem 45 times – 8-shot prompts at three exemplar-complexity levels (short/medium/long) × 15 samples each (temperature 0.7, top-p 0.95, $\leq$ 1024 tokens) – to span a range of CoT lengths, giving 900 completions per level (4500 total). Answers are graded by `\boxed{}` extraction with SYMPY equivalence; length is counted in reasoning steps (non-empty lines) and tokens. Accuracy is the mean over a level's 900 samples (mean±s.e.).

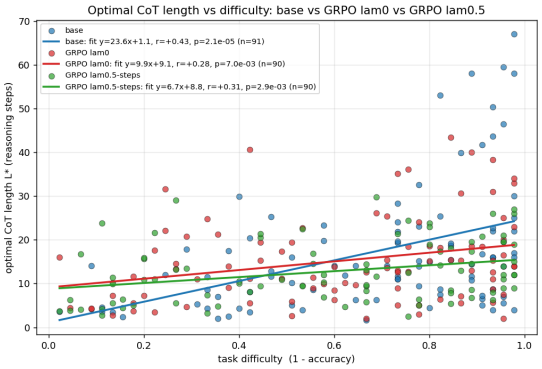
Results Our results show that our framework of length regularized GRPO outperforms pure GRPO in compressing number of steps of COT, but most importantly it produces more capable model across all difficulty levels of MATH dataset.

Table 1: Base model vs. length-regularized GRPO (GRPO+len, λ=0.5, penalty in reasoning steps) on MATH, per difficulty level.

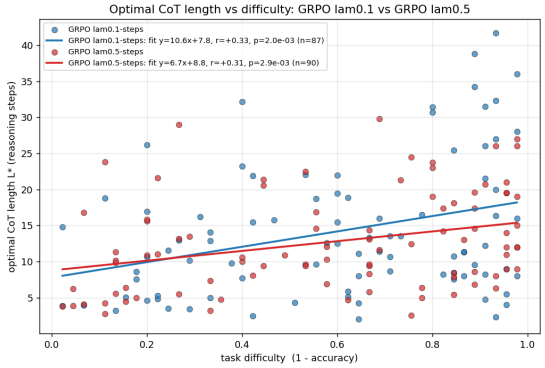
	Accuracy			Optimal CoT length (steps)		
Level	baseline	GRPO+len	Δ	baseline	GRPO+len	Δ
L1	0.470 ± 0.033	0.593 ± 0.016	+0.123	8.8	7.6	−1.2
L2	0.461 ± 0.033	0.618 ± 0.016	+0.157	12.5	11.2	−1.3
L3	0.237 ± 0.028	0.296 ± 0.015	+0.059	22.6	12.8	−9.8
L4	0.194 ± 0.026	0.287 ± 0.015	+0.093	17.7	15.1	−2.6
L5	0.089 ± 0.019	0.118 ± 0.011	+0.029	25.0	16.0	−9.0
Pooled	**0.290**	**0.382**	**+0.092**	17.2	11.2	−6.0

Table 2: Outcome-only GRPO (λ=0) vs. length-regularized GRPO (GRPO+len, λ=0.5, penalty in reasoning steps) on MATH, per difficulty level.

	Accuracy			Optimal CoT length (steps)		
Level	GRPO	GRPO+len	Δ	GRPO	GRPO+len	Δ
L1	0.512 ± 0.017	0.593 ± 0.016	+0.081	8.6	7.6	−1.0
L2	0.578 ± 0.016	0.618 ± 0.016	+0.040	12.4	11.2	−1.2
L3	0.280 ± 0.015	0.296 ± 0.015	+0.016	17.5	12.8	−4.7
L4	0.244 ± 0.014	0.287 ± 0.015	+0.043	18.3	15.1	−3.2
L5	0.106 ± 0.010	0.118 ± 0.011	+0.012	16.6	16.0	−0.6
Pooled	**0.344**	**0.382**	**+0.038**	13.2	11.2	−2.0



(a) Impact of GRPO. Optimal CoT length vs. task difficulty (1 − accuracy) for the base model and GRPO at λ=0 and λ=0.5. Both GRPO variants flatten the length–difficulty slope relative to the base model (23.6 → 9.9 and 6.8), i.e. GRPO suppresses the tendency to write longer chains on harder problems. GRPO with additional length regularization at λ=0 does so more noticeably than pure GRPO.



(b) Effect of the regularization strength λ. GRPO+len with λ=0.1 vs. λ=0.5. A stronger length penalty further reduces how steeply CoT length grows with difficulty (slope 10.6 → 6.8), confirming that λ monotonically controls length compression.

Figure 3: Optimal CoT length vs. task difficulty, with per-condition linear fits (Pearson r, p, n in each legend). One point per problem: $x = 1 - \text{accuracy}$, $y =$ mean reasoning steps of that problem's correct solutions.

References

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A A small penalty (λ=0.1) barely differs from pure GRPO

Before deciding on λ=0.5 we also trained with a weak penalty, λ=0.1. Its behaviour is almost indistinguishable from outcome-only GRPO (λ=0): pooled accuracy is essentially identical (0.344 vs. 0.348, Δ=+0.004), the per-level accuracy differences are within one standard error, and the optimal-length regression slopes are nearly the same (9.9 vs. 10.6; Fig. 4). A small length penalty adds little on top of GRPO's own simplicity bias, only the stronger λ=0.5 (main text) measurably compresses CoT length further.

Table 3: Outcome-only GRPO (λ=0) vs. weakly length-regularized GRPO (λ=0.1, penalty in reasoning steps) on MATH, per difficulty level. Differences are small and mostly within one SEM.

	Accuracy			Optimal CoT length (steps)		
Level	λ=0	λ=0.1	Δ	λ=0	λ=0.1	Δ
L1	0.512 ± 0.017	0.529 ± 0.017	+0.017	8.6	8.2	−0.4
L2	0.578 ± 0.016	0.547 ± 0.017	−0.031	12.4	12.2	−0.2
L3	0.280 ± 0.015	0.283 ± 0.015	+0.003	17.5	17.9	+0.4
L4	0.244 ± 0.014	0.262 ± 0.015	+0.018	18.3	14.2	−4.1
L5	0.106 ± 0.010	0.117 ± 0.011	+0.011	16.6	20.7	+4.1
Pooled	**0.344**	**0.348**	**+0.004**	13.2	14.2	+1.0

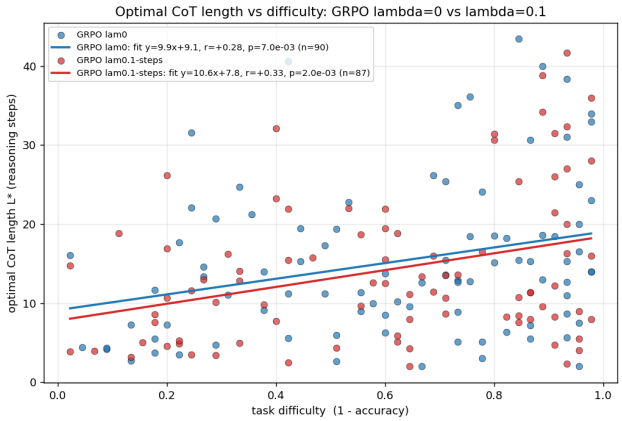
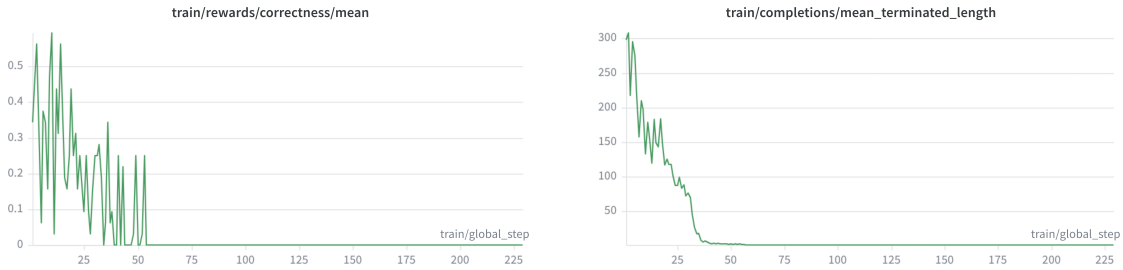


Figure 4: Optimal CoT length vs. task difficulty for GRPO λ=0 and λ=0.1. The two fits nearly coincide, confirming that a small penalty does not meaningfully change the length–difficulty relationship.

B Vanilla reward exhibits inherent training instability

We include training plots from initial phase of development, where we used the initial formulation of the reward. As we can see, the model collapses, producing empty rollouts.



C Wu et al. (2025) results extend to smaller (0.5B) models

Wu et al. (2025) report the optimal CoT length $L^\star$ only for models of 1.5B parameters and larger, where $L^\star$ decreases monotonically with model size (from ∼14 steps at 1.5B down to ∼4 steps at 72B). We extend this observation below the paper's range, to Qwen2.5-0.5B. Using the same

approach (accuracy as a function of realized reasoning steps, with $L^\star$ taken as the peak-accuracy length), the 0.5B model exhibits the same inverted-U shape and an optimum at $L^\star \approx 22$ steps. This is consistent with the paper's trend: as the model size decreases, the optimal reasoning length grows (4 at 72B $\rightarrow$ 14 at 1.5B $\rightarrow \approx 22$ at 0.5B).

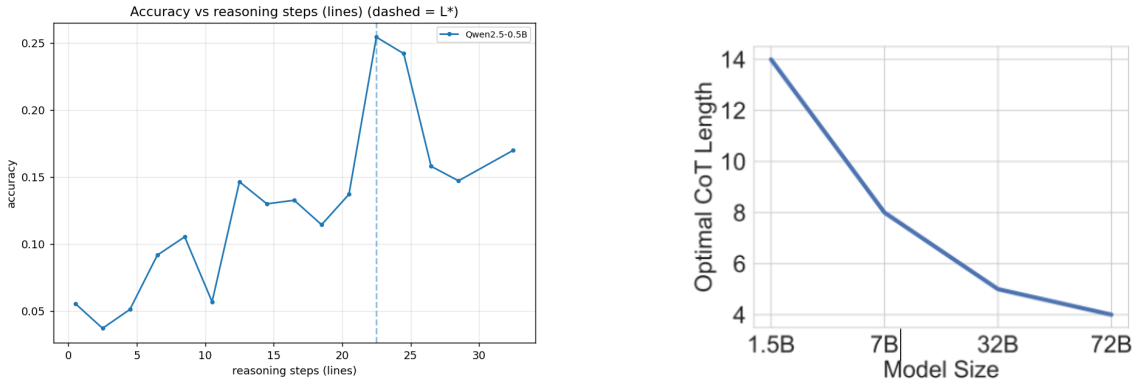


Figure 6: Extending the optimal-CoT-length trend to 0.5B. **Left:** accuracy vs. number of reasoning steps for Qwen2.5-0.5B; the dashed line marks the peak-accuracy optimum $L^\star \approx 22$ steps. **Right:** optimal CoT length vs. model size as reported by Wu et al. (2025) (1.5B–72B). The 0.5B optimum (≈ 22 steps) sits above the 1.5B value (14), continuing the inverse relationship between model size and optimal reasoning length below the paper's smallest model.